DocShield AI
AI-Powered Document Forgery Detection & Verification System

1. Introduction

DocShield AI is a Java-based web application that verifies the authenticity of digital documents using OCR, computer vision, and AI. It helps institutions detect forged certificates, IDs, invoices, and other documents while reducing manual verification effort.

2. Problem Statement

Manual verification is slow, error-prone, and often fails to detect sophisticated digital edits. An automated intelligent solution is required.

3. Objectives

• Verify uploaded documents automatically.
• Extract text using Tess4J OCR.
• Detect tampering using OpenCV.
• Analyze metadata and QR codes.
• Generate an authenticity score and downloadable report.

4. Technology Stack

Frontend: HTML, CSS, JavaScript
Backend: Java Spring Boot, Spring MVC, Spring Data JPA
Database: MySQL
AI & Libraries: Tess4J, OpenCV Java, ZXing, Apache PDFBox, Metadata Extractor, Deep Java Library (DJL)
Build Tool: Maven

5. System Architecture

User → HTML/CSS/JavaScript → Spring Boot → OCR (Tess4J) + OpenCV Analysis + Metadata + QR Verification → Authenticity Scoring → MySQL → PDF Report.

6. Workflow

1. Upload document.
2. Validate file.
3. OCR text extraction.
4. Metadata analysis.
5. Image tampering detection.
6. QR verification.
7. Calculate authenticity score.
8. Generate verification report.

7. Key Features

User authentication, document upload, OCR extraction, metadata analysis, QR verification, image tampering detection, authenticity score, report generation, verification history.

8. Applications

Educational institutions, HR recruitment, banking, government departments, insurance, legal verification, digital forensics.

9. Future Scope

Blockchain verification, mobile application, face verification, multilingual OCR, institution API integration.

10. Conclusion

DocShield AI provides a scalable, open-source, Java-centric solution for intelligent document verification using AI and computer vision.